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1. Introduction

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2. Stakeholders and Design Concerns

2.1. Stakeholders

Stakeholder	Responsibilities	Design Concerns
Software User	- Monitor audio input device.\newline- Monitor MIDI input device.\newline- Read audio files\newline- Play audio to output device.\newline- Play MIDI to output device.	DC-01\newline DC-02\newline DC-03\newline DC-04\newline DC-05\newline DC-06
Third-Party Developer	- Include SDK in C++ software application.\newline- Handle incoming MIDI messages.\newline- Process audio input.\newline- Route processed audio to output device.\newline- Manage audio in multiple tracks.	DC-07\newline DC-08\newline DC-09\newline DC-10\newline DC-11\newline DC-12\newline DC-13\newline DC-14\newline DC-17\newline DC-18
Maintainer	- Maintain CI/CD pipeline.\newline- Manage code repo.\newline- Manage software releases.	DC-13\newline DC-14\newline DC-15\newline DC-16\newline DC-17
Hardware	- Run on a Windows desktop.\newline- Run on an embedded Linux platform.	DC-13\newline DC-14

2.2. Design Concerns

ID	Description	Relevant Views
DC-01	Monitor audio input device.	Logical
DC-02	Monitor MIDI input device.	Logical
DC-03	Open and read WAV audio files.	Logical
DC-04	Open and read MIDI files.	Logical
DC-05	Route audio to output device.	Logical

ID	Description	Relevant Views
DC-06	Route MIDI to output device.	Logical
DC-07	Processing incoming MIDI messages.	Logical
DC-08	Processing incoming audio streams.	Logical
DC-09	Manage multiple audio tracks.	Logical
DC-10	Add one audio or MIDI input to a track.	Logical
DC-11	Attach one audio or MIDI output to a track.	Logical
DC-12	Chain multiple audio processors in one track.	Logical
DC-13	Build software SDK on Windows and Linux.	Context
DC-14	Build software SDK on x86_64 and ARM64 platforms.	
DC-15	CI/CD pipeline builds and packages software on all compatible platforms.	
DC-16	Complete unit testing and code coverage.	
DC-17	Package software as an SDK used by third-party software.	Context, Composition
DC-18	Third-party software developers manage audio tracks, system audio/MIDI devices, and filesystem.	Composition

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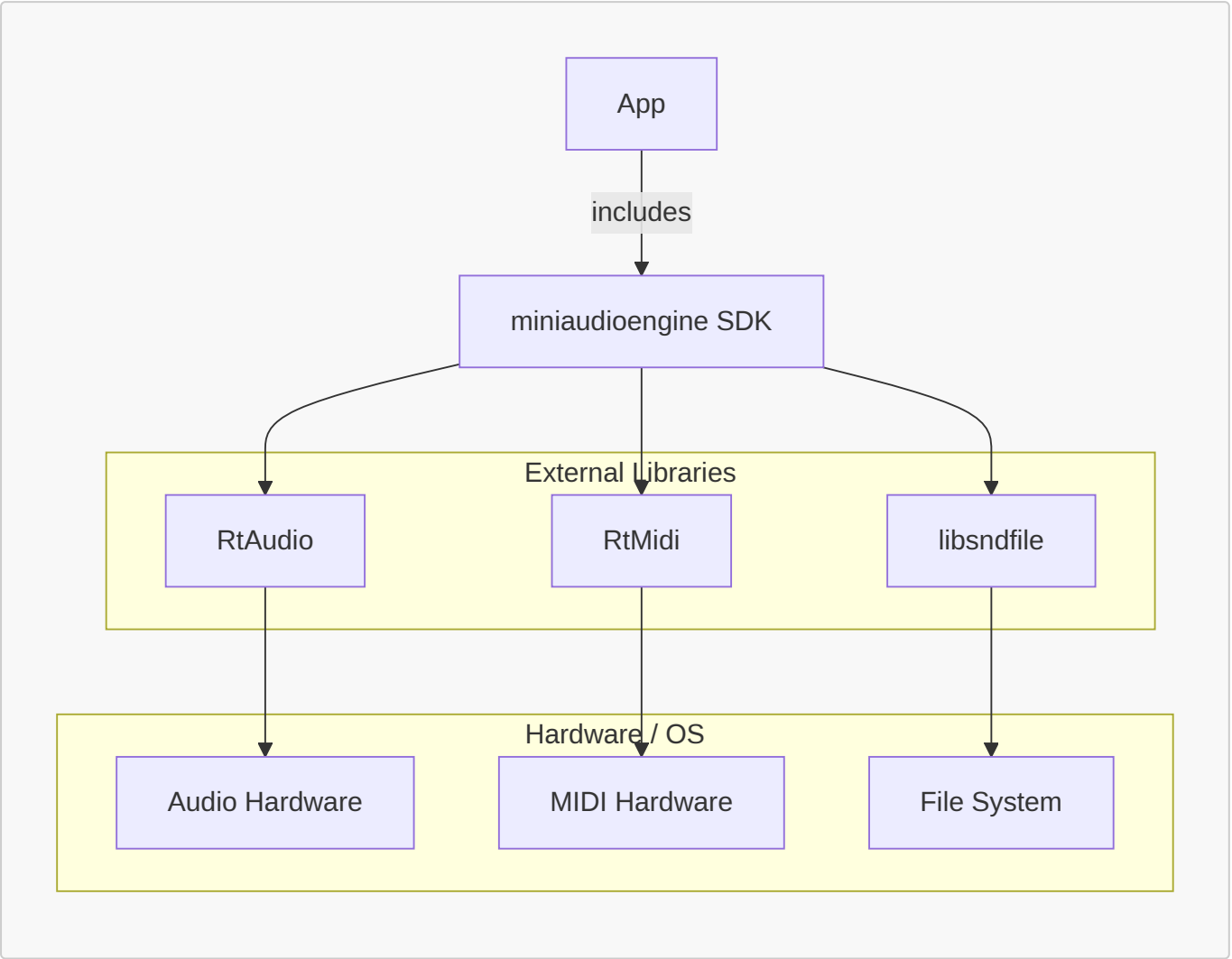
3. Design Views

3.1. Context View

Describes the software in context with its external environment. Define users, external components, and the interactions between.

Design Concern	
DC-13	Build software SDK on Windows and Linux.

Design Concern	
DC-17	Package software as an SDK used by third-party software.



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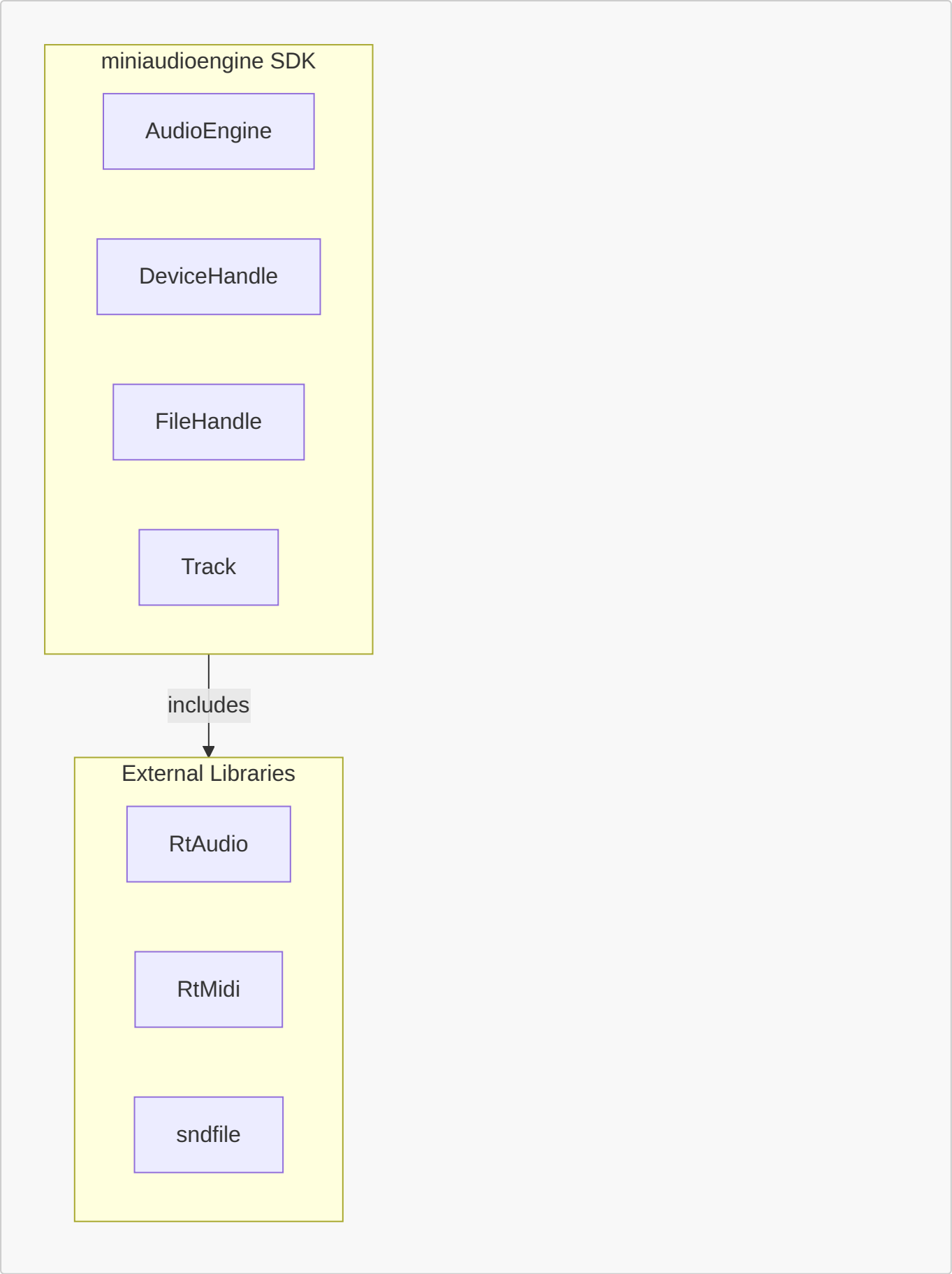
3.2. Composition View

Describe the composition of the **miniaudioengine** SDK software libraries.

Design Concern	
DC-17	Package software as an SDK used by third-party software.
DC-18	Third-party software developers manage audio tracks, system audio/MIDI devices, and filesystem.

3.2.1. miniaudioengine SDK



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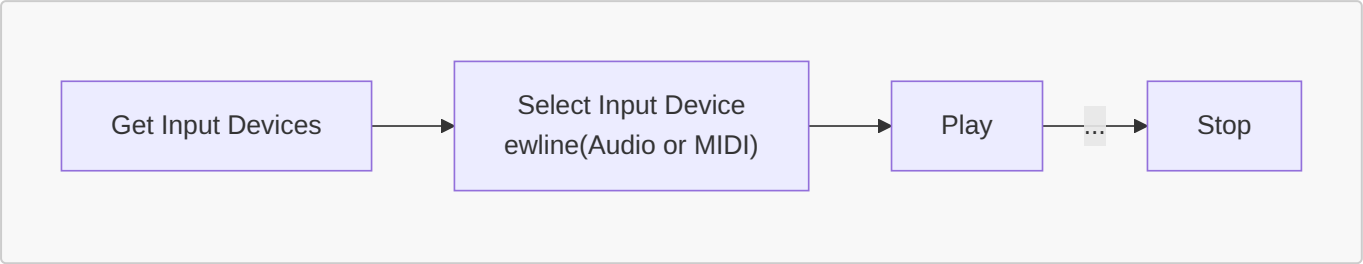
3.3. Logical View

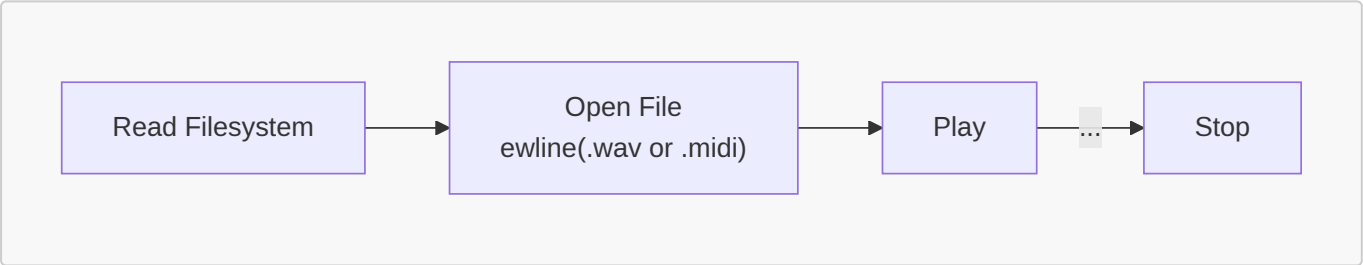
3.3.1. Record Audio Input

Design Concern	
DC-01	Monitor audio input device.
DC-02	Monitor MIDI input device.



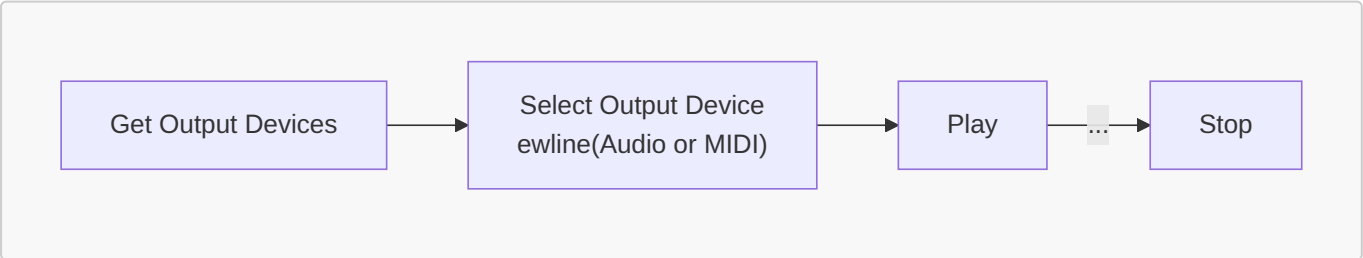
3.3.2. Play Audio File

Design Concern	
DC-03	Open and read WAV audio files.
DC-04	Open and read MIDI files.



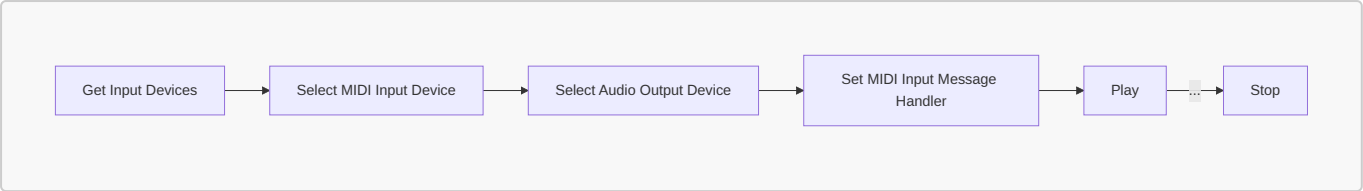
3.3.3. Play to Audio Output Device

Design Concern	
DC-05	Route audio to output device.
DC-06	Route MIDI to output device.



3.3.4. Read MIDI Messages

Design Concern	
DC-07	Processing incoming MIDI messages.



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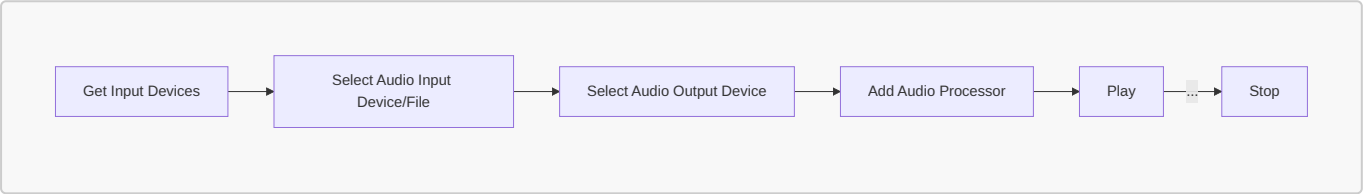
3.3.5. Audio Processing

Design Concern	
DC-08	Processing incoming audio streams.



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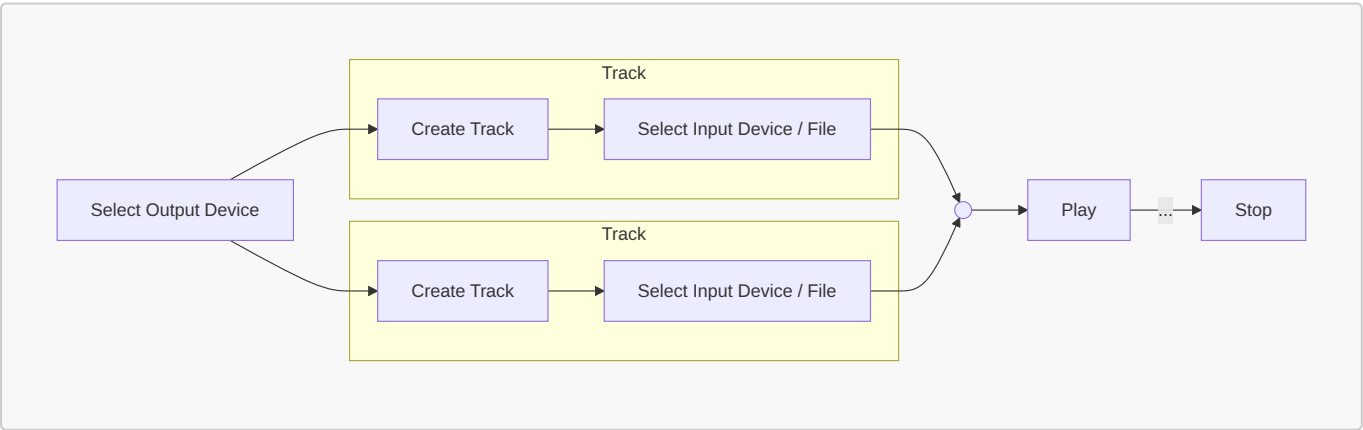
3.3.6. Multiple Tracks

Design Concern	
DC-09	Manage multiple audio tracks.
DC-10	Add one audio or MIDI input to a track.
DC-11	Attach one audio or MIDI output to a track.
DC-12	Chain multiple audio processors in one track.



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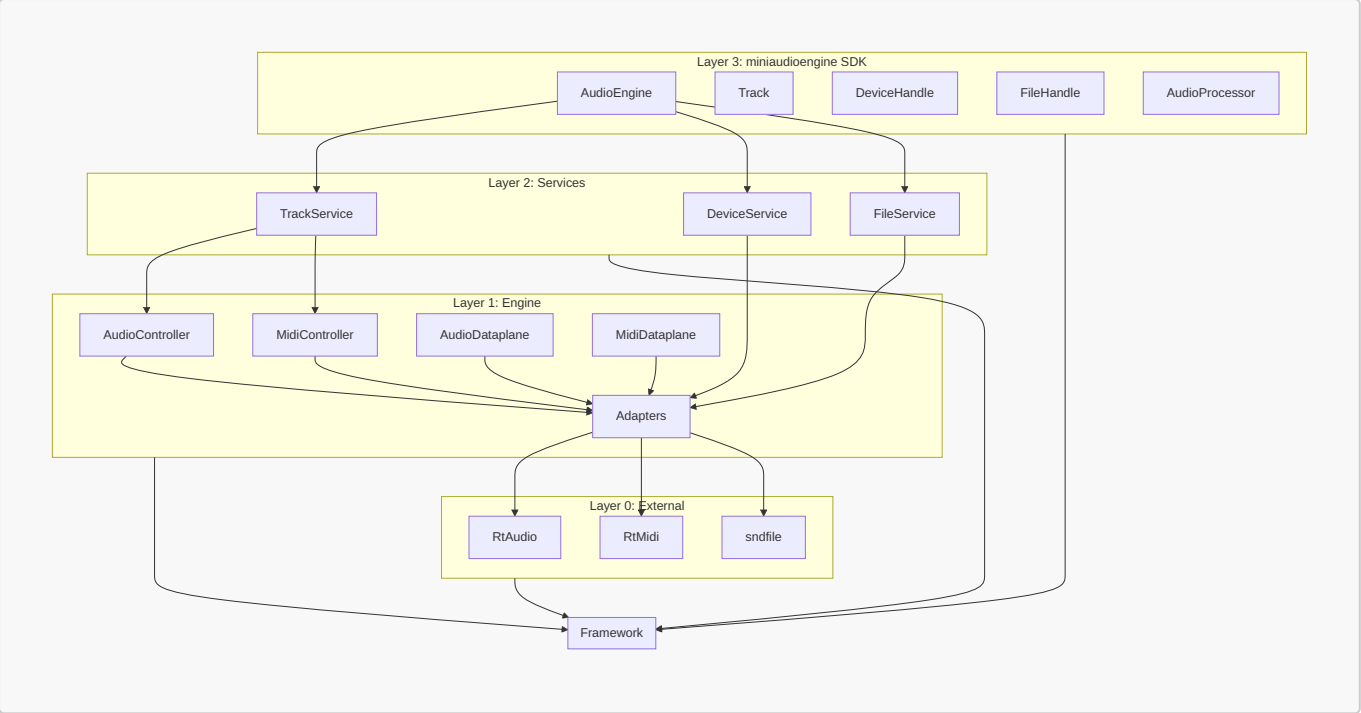
3.4. Dependency View

The components in this SDK depend on the file system, audio and MIDI devices on the host system. Separating devices and files into different services divides the dependency.

3.4.1. Layer Hierarchy



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3.5. Information View

3.5.1. Devices

3.5.2. Files

3.5.3. Tracks

Syntax error in text

mermaid version 11.14.0

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3.6. Interface View

3.6.1. Public SDK

The user uses the software via the main `miniaudioengine` SDK library.

The following types need to be accessible to the user:\newline

- `DeviceHandle`\newline
- `FileHandle`\newline
- `Track`\newline
- `Processor`\newline

The following operations need to be accessible to the user:\newline

- Get audio/MIDI devices.\newline
- Get audio/MIDI files.\newline
- Set audio/MIDI device as input or output.\newline
- Set audio/MIDI file as input or output.\newline
- Add a new track.
- Add and audio or MIDI processor to a track.\newline
- Start/stop playback.\newline
- Start/stop recording.\newline
- Start/stop monitoring.\newline

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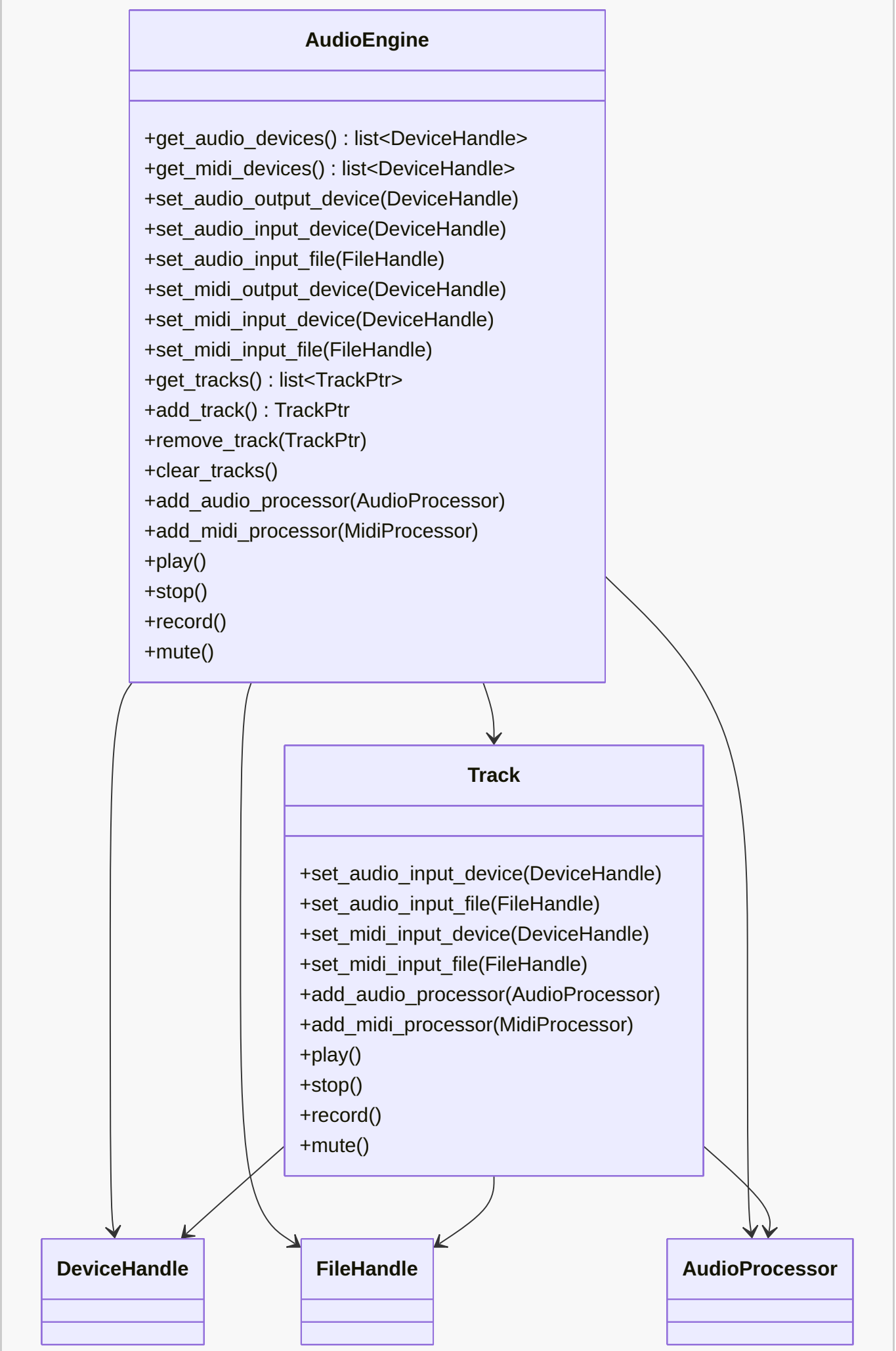
AudioEngine
+get_audio_devices() : list<DeviceHandle>
+get_midi_devices() : list<DeviceHandle>
+set_audio_output_device(DeviceHandle)
+set_audio_input_device(DeviceHandle)
+set_audio_input_file(FileHandle)
+set_midi_output_device(DeviceHandle)
+set_midi_input_device(DeviceHandle)
+set_midi_input_file(FileHandle)
+get_tracks() : list<TrackPtr>
+add_track() : TrackPtr
+remove_track(TrackPtr)
+clear_tracks()
+add_audio_processor(AudioProcessor)
+add_midi_processor(MidiProcessor)
+play()
+stop()
+record()
+mute()
Track
+set_audio_input_device(DeviceHandle)
+set_audio_input_file(FileHandle)
+set_midi_input_device(DeviceHandle)
+set_midi_input_file(FileHandle)
+add_audio_processor(AudioProcessor)
+add_midi_processor(MidiProcessor)
+play()
+stop()
+record()
+mute()
DeviceHandle
FileHandle
AudioProcessor

3.7. Interaction View

3.7.1.

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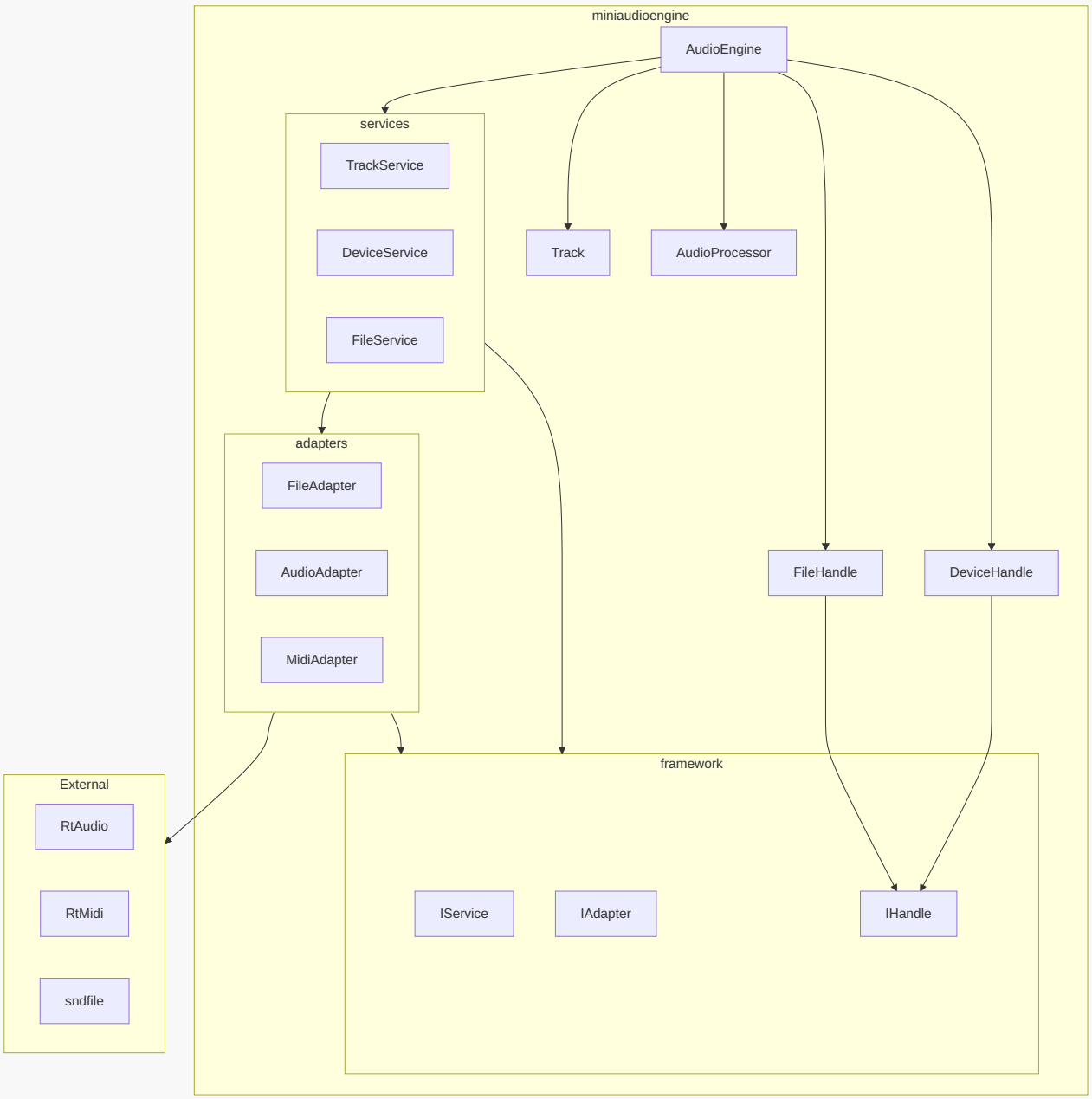
3.8. Structure View

3.8.1. Library Structure

Library	Description
framework	
services	
adapters	
RtAudio	
RtMidi	
sndfile	

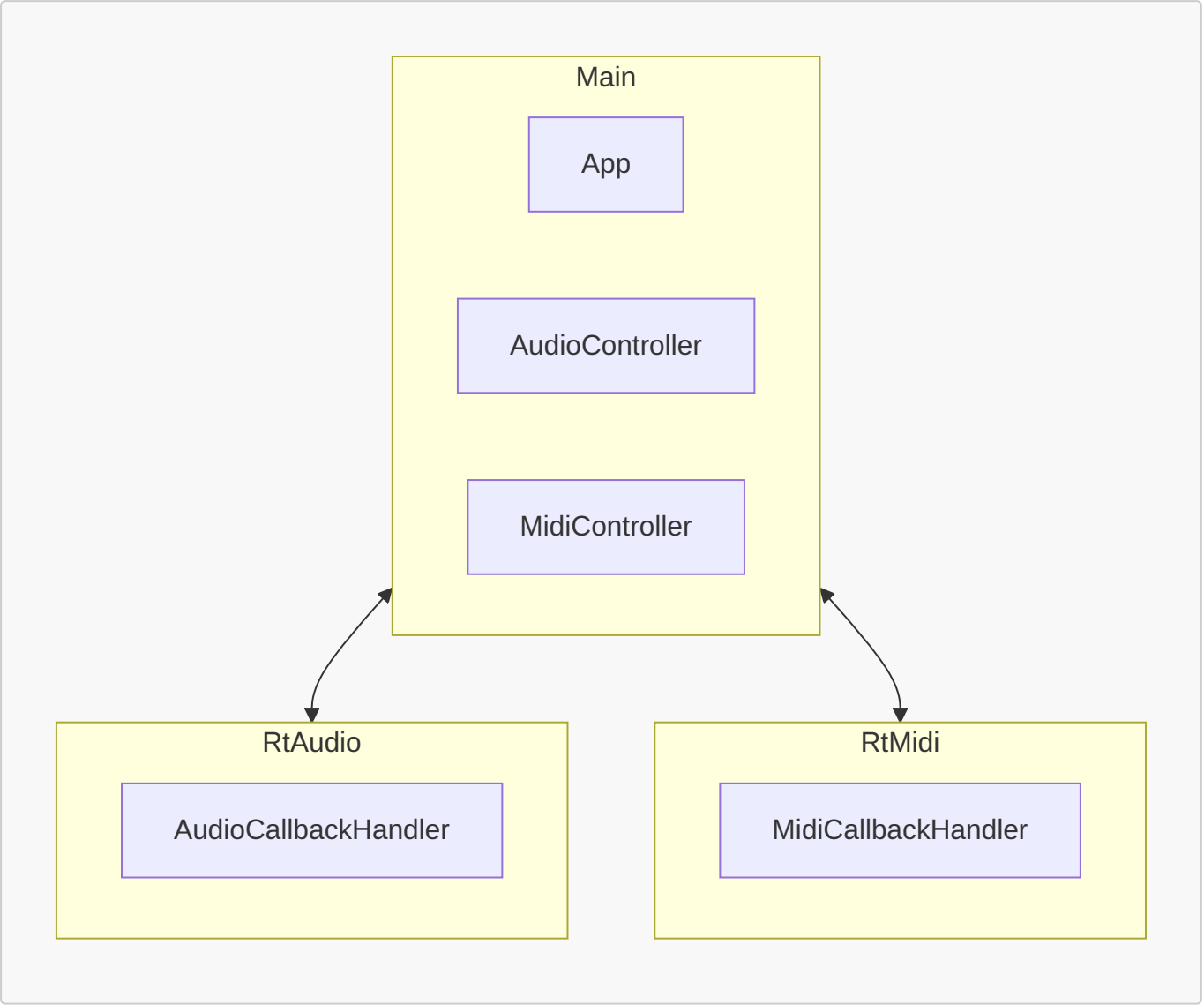


3.8.2. Project Structure

```
examples/                   # Example programs using miniaudioengine SDK
samples/
include/
    miniaudioengine/
        miniaudioengine.h   # Public facing SDK
src/
    framework/
    services/
    adapters/
tests/
```

3.8.3. Thread Structure

Thread	Description
Main	Main execution thread.
RtAudio	Real-time audio data processing.
RtMidi	Real-time MIDI message handling.



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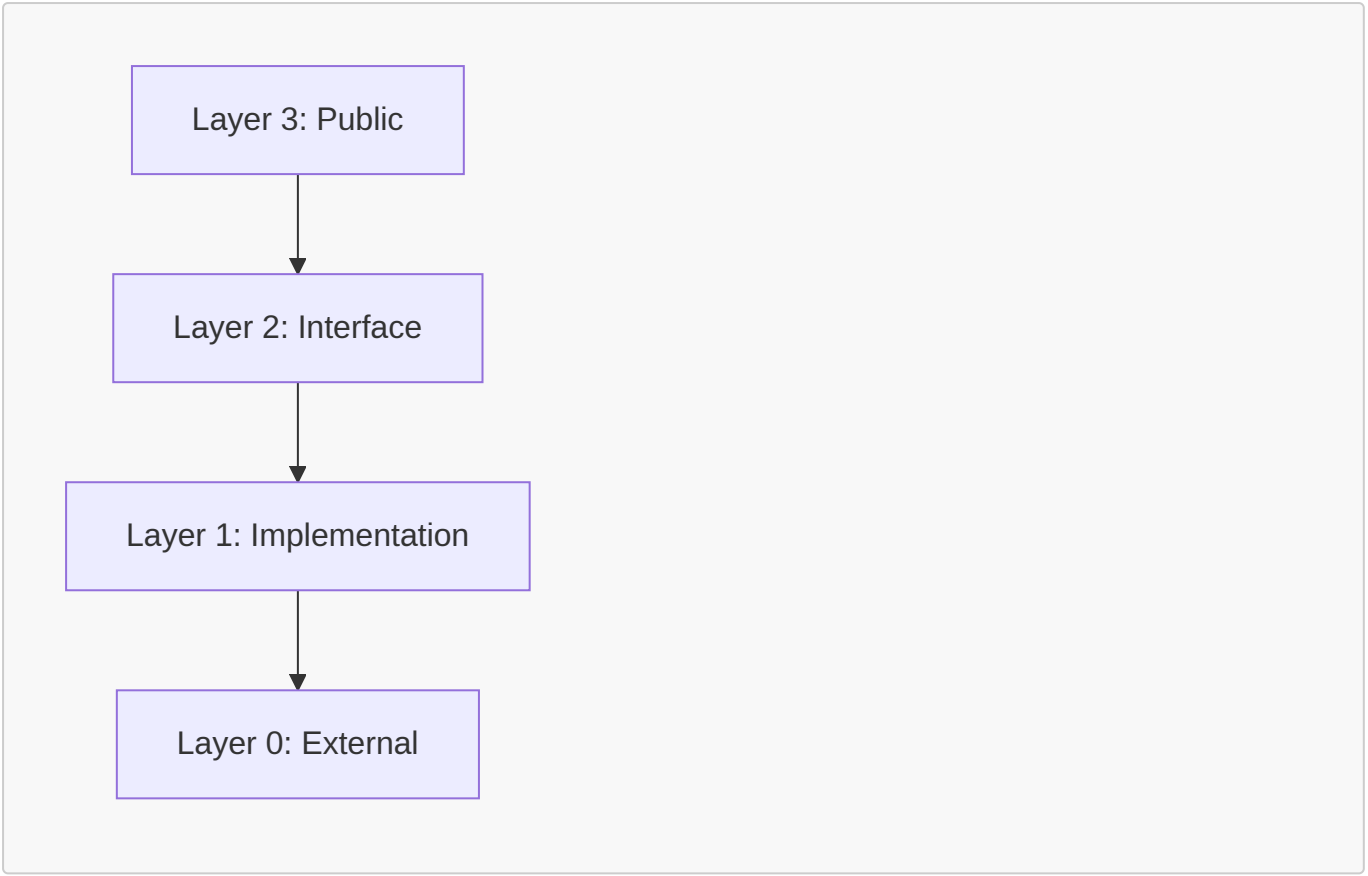
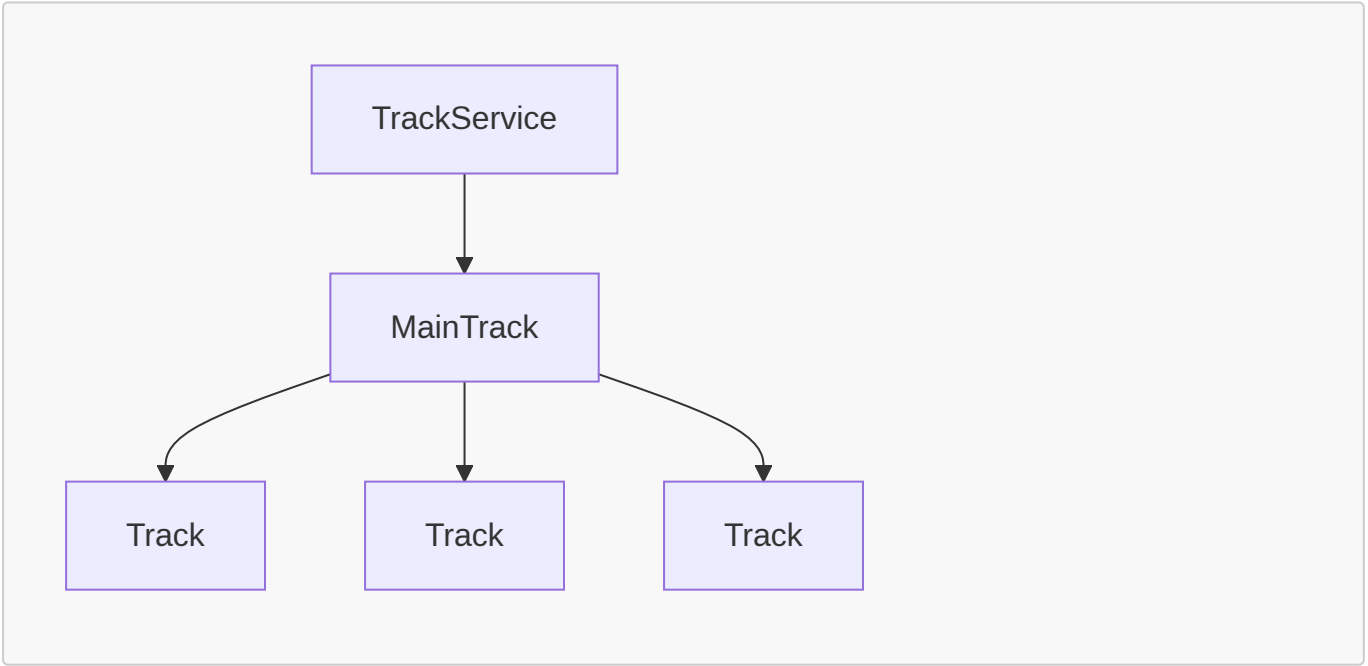
4. Design Rationale

4.1. Architectural Design

4.1.1. Layered Architecture



4.1.2. Track Hierarchy

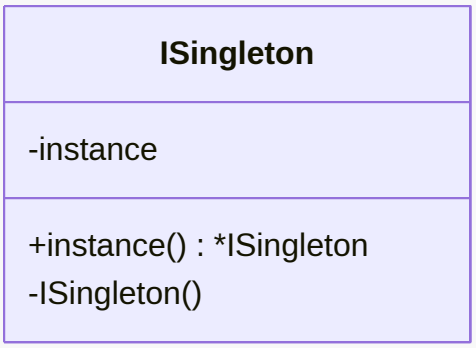


4.1.3. Software Design Patterns

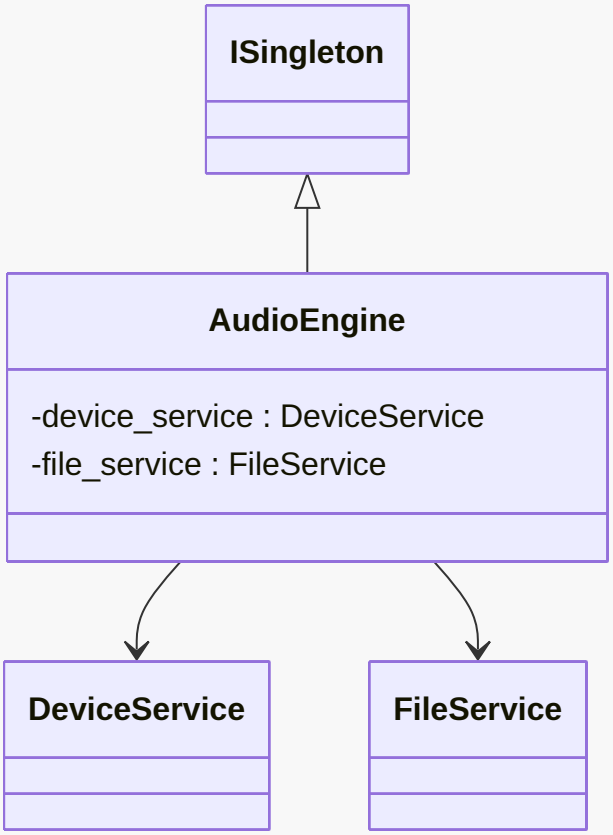
C++ PImpl

Singleton

This SDK uses the Singleton pattern for the `AudioEngine` component. There should only ever be one instance of `AudioEngine`. It owns all other components used in this SDK.



AudioEngine:

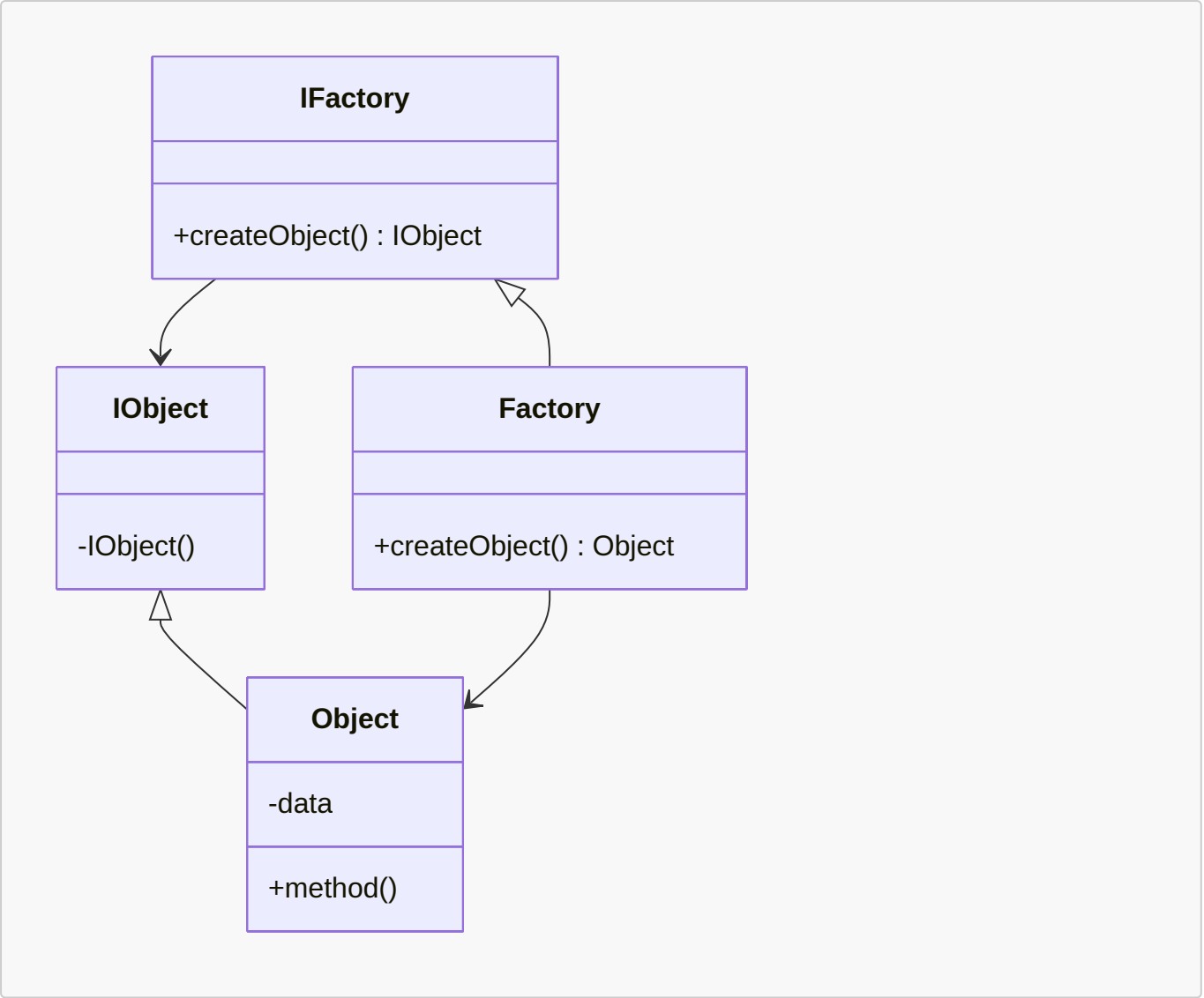


Facade

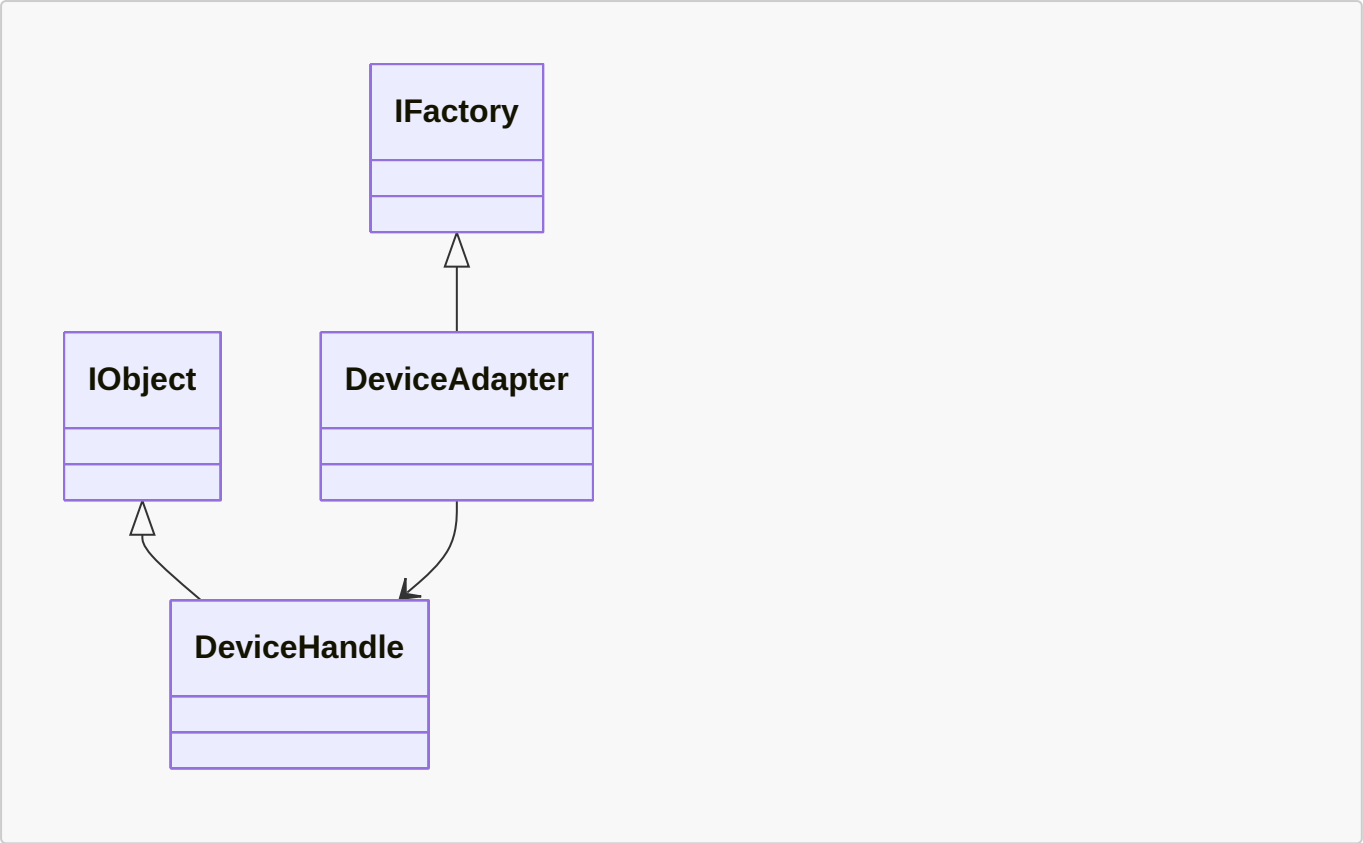
Adapter

Factory

This SDK uses the Factory pattern to create `DeviceHandle`, `FileHandle`, and `Track` objects.


IFactory
+createObject() : IObject
IObject
-IObject()
Factory
+createObject() : Object
Object
-data
+method()

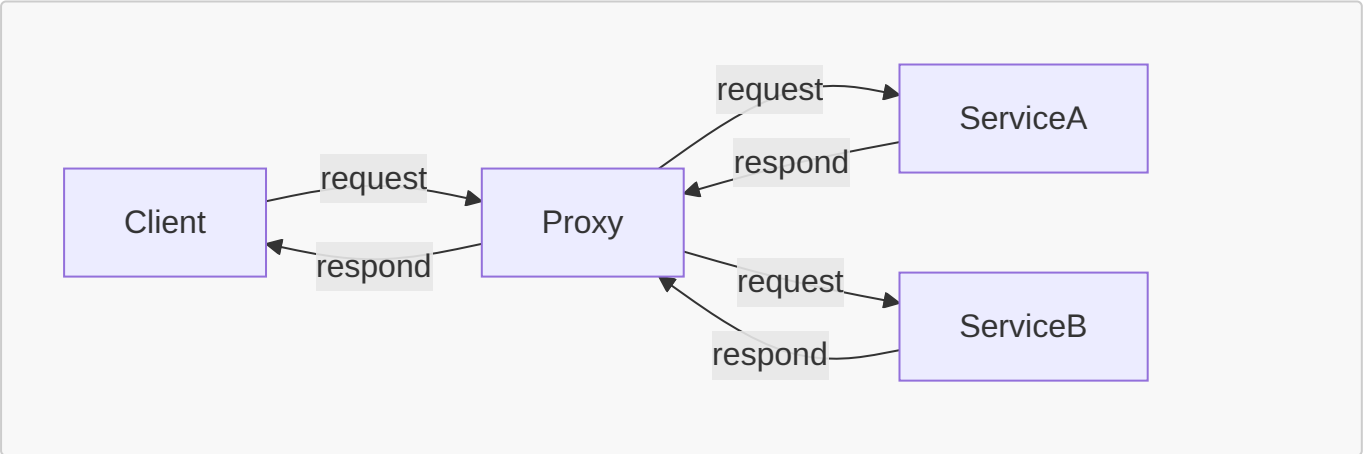
e.g.



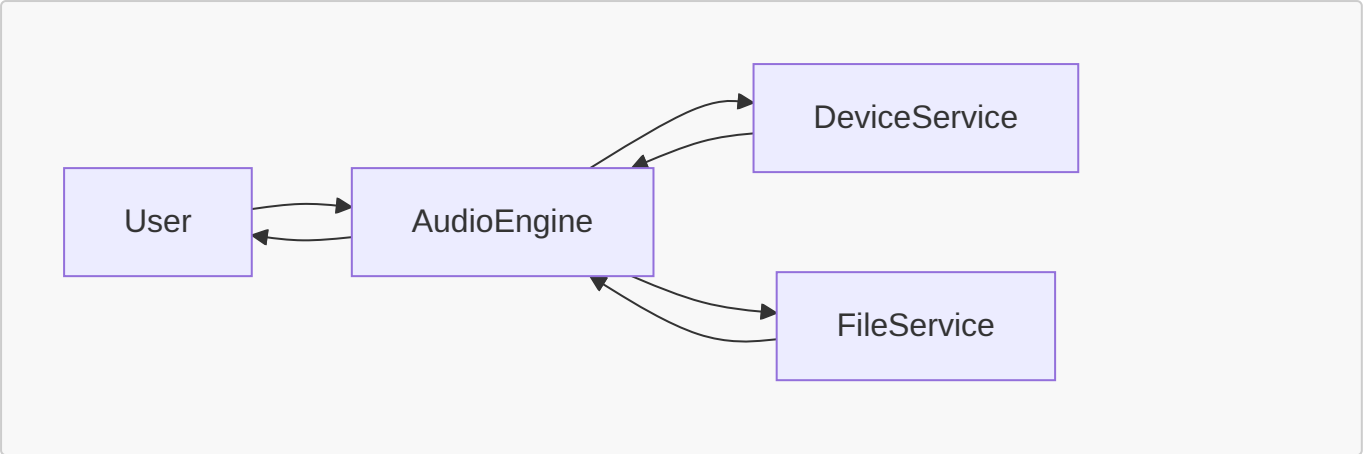
Proxy

This SDK uses the Proxy pattern. The `User` interacts with the `DeviceService` and `FileService` via the `AudioEngine`

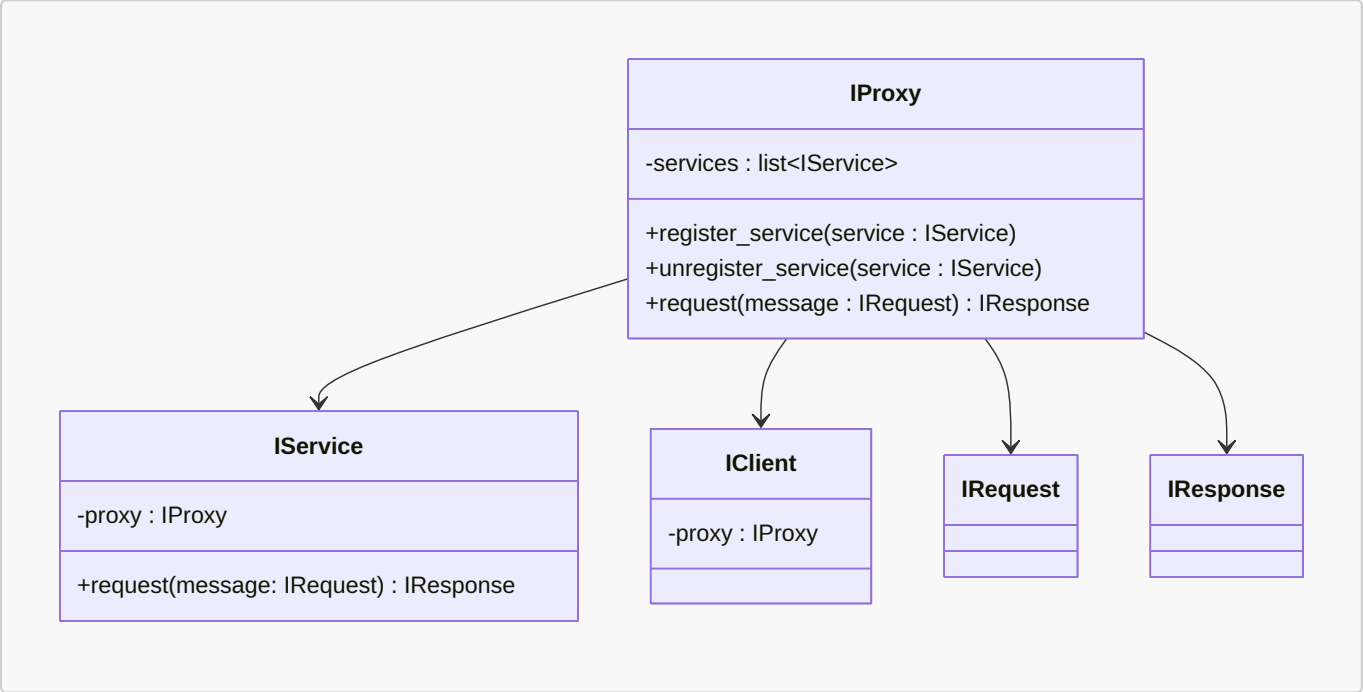
Client requests data from a Service via a Proxy.



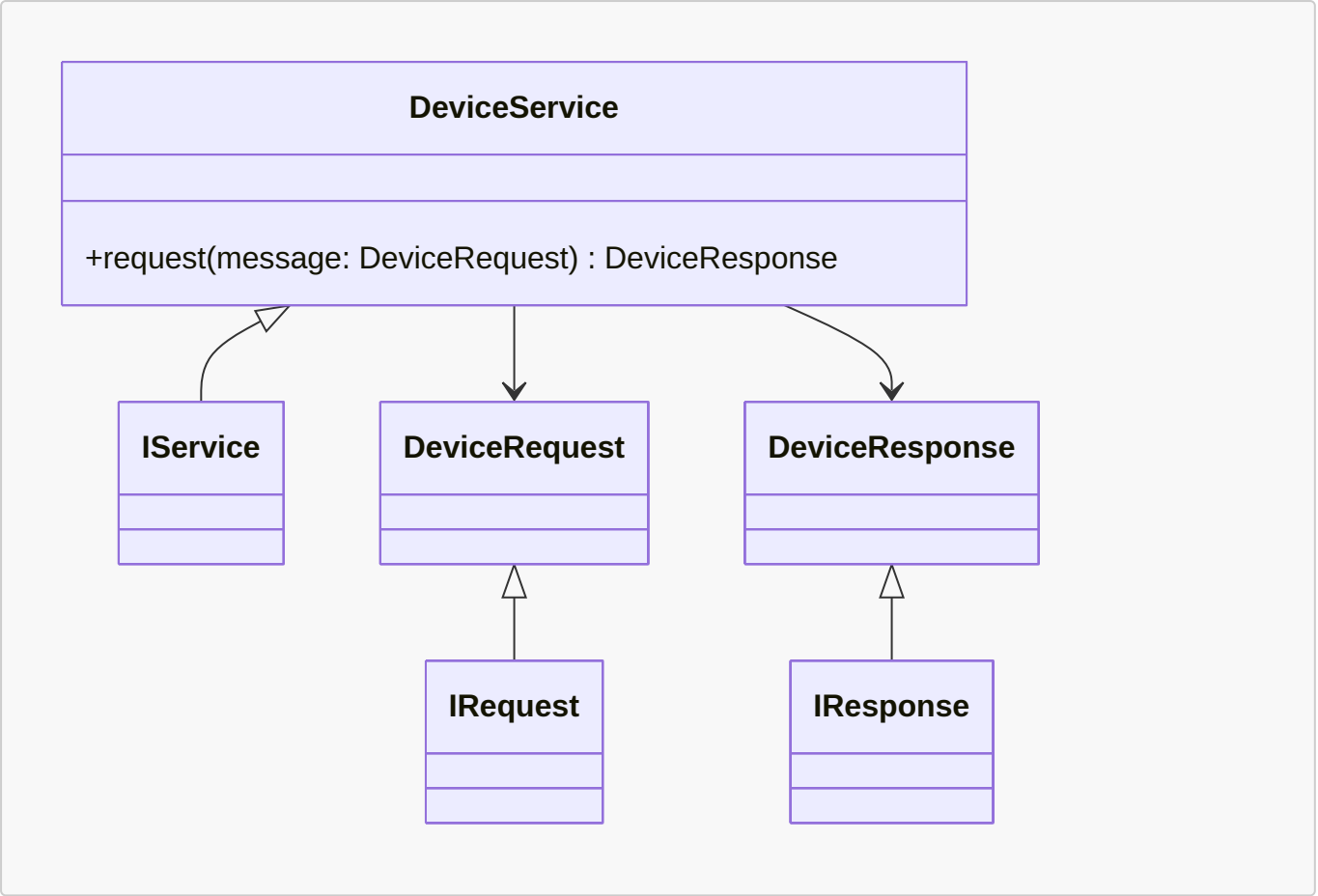
e.g.



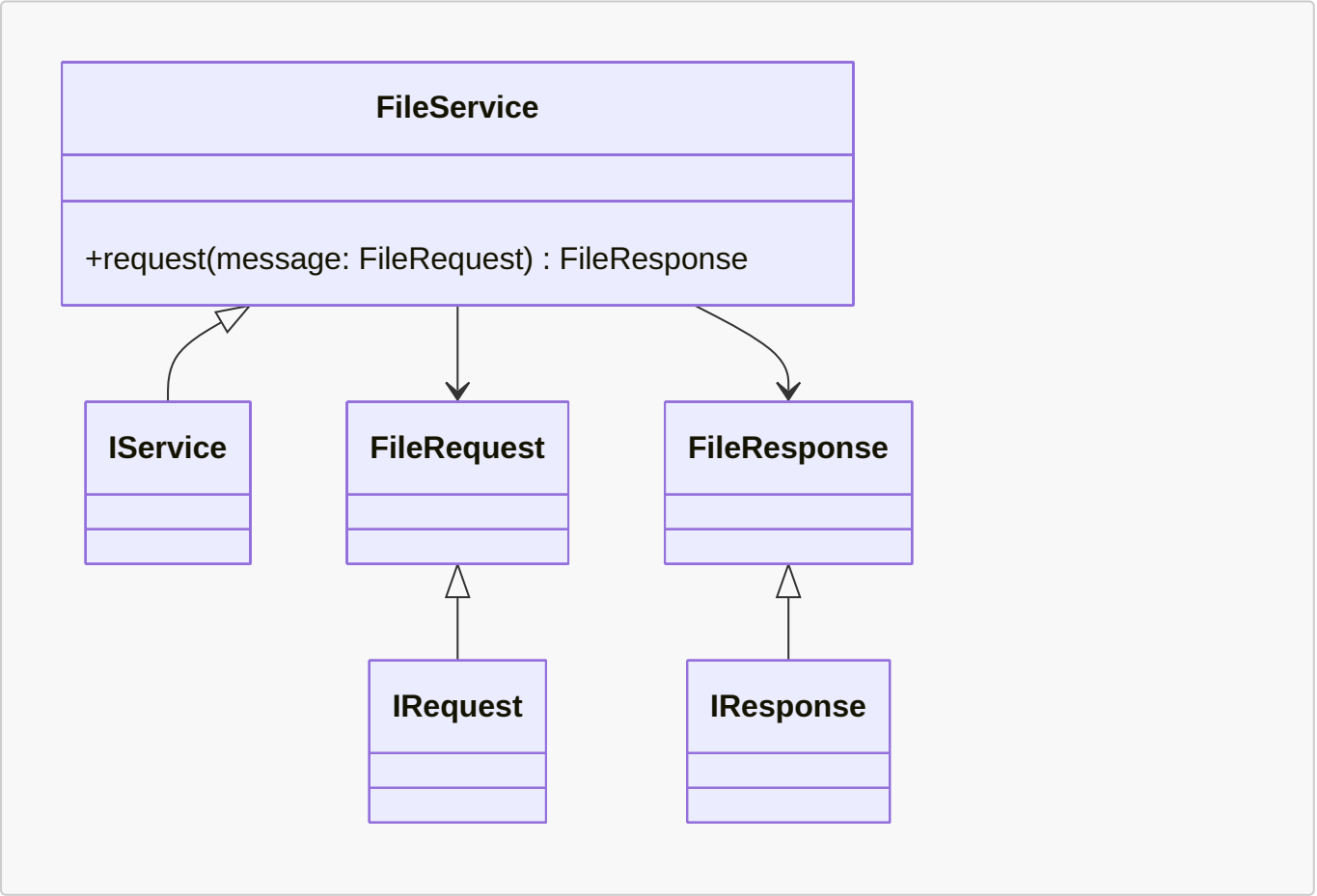
Note: The request method in this example is blocking.



Device Service:



File Service:



4.2. External Libraries